



THE NATIONAL WATER MODEL

This water prediction model creates forecast guidance for over 3.4 million miles of rivers and streams across the United States and its territories. The NWM also guides NWS field offices to support the delivery of expanded water services, as part of NOAA's integrated environmental intelligence and prediction capabilities.



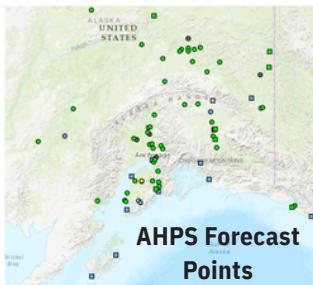
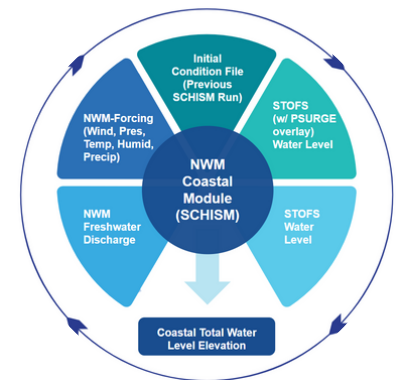
Image: National Water Model Streamflow output points.

VERSION 3.0 OF THE NWM

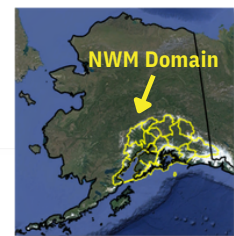
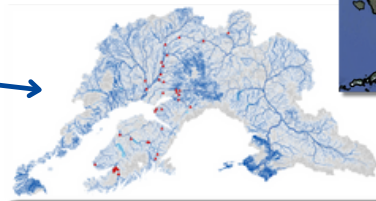
With an implementation date of September 2023, V3.0 features several significant upgrades that emerged from the development efforts of the joint OWP-NCAR (National Center for Atmospheric Research) development team.

HIGHLIGHTS

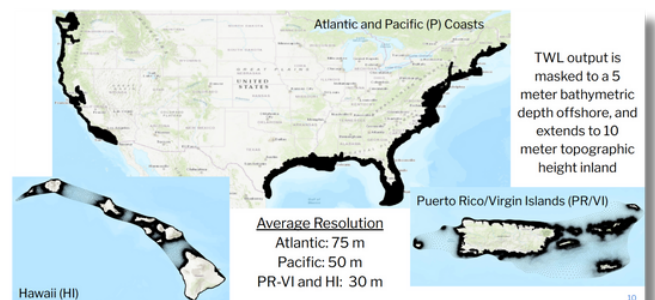
- First time provision of **NWM Total Water Level guidance for coastal areas** of the Continental United States (CONUS), Hawaii and Puerto Rico / US Virgin Island domains. This is accomplished via use of the Semi-implicit Cross-scale Hydrosphere Integrated System Model (SCHISM) integrated within the NWM, to couple NWM freshwater discharge estimates with oceanic forcing from the Surge and Tide Operational Forecast System (STOFS) and Probabilistic Tropical Storm Surge (P-SURGE) model.
- NWM Domain expansion to south-central Alaska (Cook Inlet, Copper River Basin, and Prince William Sound regions), enabling provision of NWM operational hydrologic model forecast guidance to this region.
- Use of Multi-Radar Multi-Sensor (MRMS) precipitation as forcing for the NWM Analysis and Assimilation configuration over the Puerto Rico / US Virgin Island domain.
- Addition of the National Blend of Models (NBM) as a forcing source for NWM CONUS medium-range forecasts and Alaska short-range and medium-range forecasts.



Dense network of NWM hydrologic guidance



- Ingest of RFC-supplied reservoir outflow forecasts at 77 additional locations, bringing the total of such sites to 392.
- Enhancements to the treatment of reservoirs, land surface parameters and calibration/regionalization approach leading to improvements in model skill.
- Implementation of a new surface runoff scheme (Xinanjiang) within the NWM, improving streamflow simulation performance.



NEXTGEN FRAMEWORK



OWP | OFFICE OF
WATER
PREDICTION

Improvements in NWM performance from version to version, by calibration, are beginning to plateau. Whereas the current NWM applies the same formulation everywhere, the hydrologic literature suggests improvements can be made through use of specific model formulations tailored regionally for individual streamflow generation processes. The Office of Water Prediction (OWP) proposed the development of an interagency "Next Generation Water Resources Modeling Framework" or NextGen. The Next Generation National Water Model (Nextgen NWM) will represent a particular configuration of the NextGen framework. The NextGen framework promotes model interoperability, standardizes data and model setup workflows, and eases implementation and evaluation of different modeling approaches. The potential of the NextGen framework has led OWP to work with other federal water science and modeling partners, including the U.S. Army Corps of Engineers, U.S. Geological Survey, and the U.S. Bureau of Reclamation, as well as the academic research community.

An interagency requirements meeting led to the following set of "design tenets" in establishing the NextGen framework:

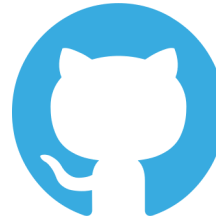
- Maximum flexibility (model agnostic)
- Common architecture to promote interoperability
- Open source and standards-based development
- Friendly to domain scientists and engineers to facilitate community development
- Two-week target to allow graduate students or new employees to add functionality supported by clear and concise documentation and step-by-step examples/tutorials and training

A second interagency meeting produced agreement on a two-pronged approach for model coupling standards. The first uses the existing Basic Model Interface (BMI) standard for model coupling, and extends it to allow more efficient use of models on supercomputers. The second involves developing a component library of standard flux calculators, integrators, and numerical solvers, to enable the framework to apply standard methods for calculation across the many potential hydrological models deployed in the framework.

The NextGen framework is in the early stages of development and OWP expects that the NextGen prototype will serve as the starting point for development. Together with our federal partners, academic and research centers, the NextGen framework offers the potential to create a powerful research, development, testing, and operational system to advance hydrologic and water resources prediction at temporal scales ranging from sub-hourly to decadal, and spatial scales ranging from catchments to continents.

OWP will identify the features included in the NextGen configuration that replaces the current operational NWM in 2024. The reliance on open source development practices, data and model coupling standards enhances interoperability between federal agencies, speeds the adoption of advances from the research community, and creates a new and exciting opportunity to stop reinventing models, and a framework to advance successful formulations.

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The Next Generation Water Resources Modeling Framework promotes model interoperability on a common data model through use of established geoscience standards. Implementing standards-based development in an open-source environment is transformative for hydrologic prediction; it advances model developers beyond the need to invent data models and model coupling strategies. This new framework offers unprecedented opportunities for collaboration with federal partners and the broader research community to enhance modeling to meet water resource challenges of the 21st century.

- Fred Ogden

OWP Chief Scientist